

Santiago Giraldo Godoy

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EDUCATION

Worcester Polytechnic Institute (WPI)

Bachelor of Science in Computer Science, Minor in Robotics Engineering

May 2026

Graduated with High Distinction | GPA: 3.83/4.0 | Upsilon Pi Epsilon (UPE) Honor Society

Data Science Outstanding MQP Award (2026) – Awarded for AI for Impact: Inferring Suicidal Ideation via Smartphones

SKILLS

Programming Languages: Java, TypeScript, C++, C, JavaScript, CSS, HTML, PL/SQL, PostgreSQL, Kotlin, Python, Swift

AI / Machine Learning: PyTorch, Deep Learning, CNNs, Machine Learning Model Training, Prompt Engineering, RAG tools, AI Agents

Software Frameworks: React.js, Shadcn UI, Express, Tailwind, Bootstrap, Node.js, JWT, MERN Stack

Software Development Tools: Webstorm, IntelliJ, VS Code, Android Studio, Jira, Matlab, R Studio, Git, GitHub, Unix command line

Languages: Spanish: Native, English: Bilingual proficiency

Development Skills: Agile Development, Scrum, Object-Oriented Design, Design Patterns

EXPERIENCE

Peer Tutor for the Academic Resource Center, WPI, Worcester

Aug 2023 – Mat 2026

- Supported students in becoming independent learners by helping them comprehend lecture content and practice problems, using skills in C++ and Java to break down complex concepts.
- Reinforced concepts from 10 courses, including calculus and computer science, during tutoring sessions, resulting in **25% of students returning** for additional support.

Residential Advisor for Housing and Residential Services, WPI, Worcester

Aug 2024 – May 2026

- Fostered a positive and inclusive community by organizing events, mediating conflicts, and providing guidance to residents.
- Ensured safety and policy compliance by responding to crises, and documenting incidents during on-call shifts.
- Collaborated with Residential Services staff to plan programs, facilitate move-ins and move-outs, and assist with administrative tasks such as health and safety checks and maintenance reports.

Software Engineer Intern, Oracle, Redwood Shores

May 2025 – Aug 2025

- Improved AI agent response quality through **prompt engineering and optimization of Retrieval-Augmented Generation (RAG)** workflows, increasing **human-like response rates by 50%**.
- Migrated and optimized data unification from multiple sources into a single, standardized database through SQL and Java endpoints allowing more than a data transfer improvement **from 6 sec/entry to 0.2 sec/entry, shortening migration by 97%**.
- Created multiple Java endpoints to handle data filtration, error handling, and data storage and retrieval **causing 25% more accuracy**.
- Improved AI chatbot responses by unifying databases, improving prompt handling and allowing user feedback to be stored for review, resulting in a beta version of a troubleshooting agent meant to **reduce expert workload and resource consumption by 60%**.

AI Project worker, WPI, Worcester

Jan 2023 – Dec 2024

- Taught an AI program with over **300 images per week**, enhancing the machine vision capabilities of the AI.
- Trained AI software to safeguard 5th grade students' confidential information, ensuring data privacy and security for over 1000 students.

Chief Operating Officer, Universal Education Initiative (UEI), Worcester

Sep 2023 – Sep 2024

- Managed day-to-day activities of the organization, ensuring correct synchronization by 5 teams, which improved operational efficiency
- Developed timelines, established roles and responsibilities, and set company objectives, leading to improved company performance and goal alignment
- Tracked different company teams' performance and created biweekly reports for the CEO, which facilitated informed decision-making and strategic planning

Lead Instructor at Bentley Camp, iDTech, Waltham

Jun 2024 – Jul 2024

- Taught over 80 students in 7 week-long classes, how to design, build, program, repair, and pilot Vex V5 Battle Bots tournament.
- Managed a team of 10 instructors at the camp, ensuring smooth operations and positive experiences for over 800 students aged 7 to 17
- Handled medical emergencies in the absence of the health director, ensuring safe and effective medicine administration.

ACADEMIC PROJECTS

AI for Impact: Inferring Suicidal Ideation via Smartphones, WPI

Aug 2025 – May 2026

Recipient of WPI's 2026 Data Science Outstanding MQP Award

- Refactored and re-architected a Kotlin Multiplatform (KMP) mental health monitoring app to support up to **80% shared Android/iOS codebases**, ensuring maintainability and cross-platform parity.
- Implemented passive smartphone data collection (motion, app usage, Bluetooth, survey data) for multi-modal behavioral analysis during an active medical research trial.
- Integrated and tested Bluetooth and background data collection workflows to ensure consistent sensor data acquisition across platforms.
- Conducted exploratory data analysis and feature engineering to correlate digital behavior patterns with self-reported suicidal ideation.

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School Benchmarking application in collaboration with Osprey Software, WPI

Jan 2026 – Mar 2026

- Collaborated on a 5-person team to create a web application prototype running in the cloud for Osprey Software.
- Developed the web application using the MERN stack, MongoDB, Express, React, Node.js, with Typescript and Tailwind.
- Created a final software prototype that was a private school benchmarking web application that included secure logins, an improved user interface for collecting annual school data, a dashboard for comparative analyses against other schools in their group or region, enforced data access and privacy.

Spreadsheet application, WPI

Jan 2026 – Mar 2026

- Designed and implemented a full-stack spreadsheet application using **Java (Spring Boot)** and **React**, following the **MVC architectural pattern** to separate frontend, backend, and business logic.
- Built a **custom expression parser** using the **Shunting Yard algorithm** to convert infix formulas into evaluable expression trees, supporting arithmetic and aggregate functions (SUM, COUNT, AVE).
- Developed an extensible evaluation engine using **Composite, Factory, and Observer design patterns**, enabling dynamic formula computation and automatic propagation of updates across dependent cells.
- Implemented support for **cell and range references (e.g., A1:B3)** with dependency tracking and circular reference detection to ensure correctness and stability.
- Wrote comprehensive **JUnit** tests to validate expression evaluation, parsing logic, and edge cases, ensuring reliability and maintainability of the system.

Robotic Manipulation System, WPI

Aug 2025 – Oct 2025

- Designed and programmed a 4-DOF robotic manipulator using MATLAB to perform coordinated manipulation tasks, including trajectory execution and object interaction.
- Implemented forward and inverse kinematics to control end-effector positioning and enable precise motion in task space.
- Developed trajectory generation and velocity kinematics algorithms to produce smooth, continuous motion between target positions.
- Integrated computer vision (intrinsic & extrinsic) to detect, localize, and map objects from image space to robot coordinates.
- Built a vision-guided pick-and-place system capable of identifying objects, planning motion, and executing sorting tasks autonomously.

Digital Tools for Making and Monitoring of Hawaiian Seed Gardens, WPI

Jan 2025 – Mar 2025

- Designed and deployed AR, GIS, and web-based tools to promote rain gardens **attracting +100 participants/month than before**.
- Led user testing and iteration with community volunteers, improving usability and delivering documentation for maintenance.
- Collected and analyzed **~80 homeowner survey responses** to find motivations, barriers, and feature preferences, shaping tool design.

Traffic Sign Recognition Models, Machine Learning, WPI

Oct 2024 – Dec 2024

- Developed a machine learning model for traffic sign classification, achieving **99.50% accuracy** using a custom CNN.
- Fine-tuned pre-trained models (ResNet50, VGG16) and optimized hyperparameters for improved performance from **70% to +95%**.
- Implemented data preprocessing and augmentation techniques to enhance model generalization.

Medical App Prototype for Brigham and Women's Hospital, Software Engineering, WPI

Mar 2024 – May 2024

- Worked on a **10-person team to apply Agile** methodologies and software design patterns in the creation of a web application prototype.
- Developed the web application using React.js, Typescript, Shadcn UI, Express, Prisma ORM, and PostgreSQL that included hospital pathfinding, a map editor, a functional EMR input and more.
- Designed backend of the application through the creation of ERD and writing the Prisma schema components creating **90% of schema**.
- Served as the technical leader of the backend subgroup handling the tRPC routes to connect the frontend and backend of the app.

Computer Networks, WPI

Oct 2023 – Dec 2023

- Implemented a **TCP-based HTTP client and server** in C/C++ using **Unix sockets**, supporting HTTP/1.1 requests, response handling, and RTT measurement for performance analysis.
- Built a **reliable data transfer protocol (Alternating Bit Protocol)** with packet loss, corruption, and retransmission handling using timers and acknowledgments.
- Developed a **distributed distance vector routing algorithm** to compute shortest paths across a simulated network, handling dynamic updates and asynchronous message passing between nodes.
- Simulated real-world network conditions including **packet loss, delay, corruption, and reordering**, improving robustness of protocol implementations.
- Analyzed network performance metrics such as **latency, throughput, and convergence behavior**, gaining insight into modern network design tradeoffs.

EXTRACURRICULARS

Film and Digital Photography

2022-Present

Soccer

2022-Present

Bowling

2022-Present

SHPE Member

2022-Present

SHPE Executive

2024